BOOKLET CODE

G5

TRIPATHI STUDY ZONE
JEE MAIN| NEET| AIIMS| KVPY| BITSAT

SEPTEMBER
13

NEET 2020

Question Paper & Answer Key

INSTRUCTIONS

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on side-1 and side-2 carefully with blue/black ball point pen only.
2. The test is of 3 hours duration and Text Booklet contains 180 questions. Each question carries 4 marks. For each correct response, the candidate will get 4 marks. For each incorrect response, one mark will be deducted from the total scores. The maximum marks are 720.
3. Use Blue/Black Ball Point Pen only for writing particulars on this page/marking responses.
4. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
5. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
6. The CODE for the Booklet is G5. Make sure that the CODE printed on Side-2 of the Answer Sheet is the same as that on this Test Booklet. In case of discrepancy, the candidate should immediately report the matter to the Invigilator for replacement of both the Test Booklet and the Answer Sheet.
7. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
8. Use of white fluid for correction is NOT permissible on the Answer Sheet.
9. Each candidate must show on demand his/her Admit Card to the Invigilator.
10. No candidate, without special permission of the Superintendent or Invigilator, would leave his/her seat.
11. The candidates should not leave the examination hall without handing over their Answer Sheet to the Invigilator on duty and sign the Attendance Sheet twice. Cases where a candidate has not signed the Attendance Sheet second time will be deemed not to have handed over the Answer Sheet and dealt with as an unfair means case.
12. Use of Electronic/Manual Calculator is prohibited.
13. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the examination hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
14. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
15. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.

TRIPATHI
STUDY ZONE, NASHIK
NEET | IIT-JEE | Olympiad

Near Vihar Hotel, Shraddha Sankul Building-1,Gangapur Road, Nashik
Phone : 9112409964

BIOLOGY

01. If the distance between two consecutive base pairs is 0.34 nm and the total number of base pairs of a DNA double helix in a typical mammalian cell is 6.6×10^9 bp, then the length of the DNA is aproximately:

(1) 2.5 meters

(2) 2.2 meters

(3) 2.7 meters

(4) 2.0 meters

Ans. (2)

02. Bilaterally symmetrical and acoelomate animals are exemplified by :

(1) Platyhelminthes

(2) Aschelminhes

(3) Annelida

(4) Ctenophora

Ans. (1)

03. Match the following columns and select the **correct** option.

	Column I		**Column II**
(a)	Gregarious, polyphagous pest	(i)	*Asterias*
(b)	Adult with radial symmetry and larva with bilateral symmetry	(ii)	Scorpion
(c)	Book lungs	(iii)	*Ctenoplana*
(d)	Bioluminescence	(iv)	*Locusta*

	(a)	**(b)**	**(c)**	**(d)**
(1)	(iv)	(i)	(ii)	(iii)
(2)	(iii)	(ii)	(i)	(iv)
(3)	(ii)	(i)	(iii)	(iv)
(4)	(i)	(iii)	(ii)	(iv)

Ans. (1)

04. Which is the important site of formation of glycoproteins and glycolipids in eukaryotic cells ?

(1) Peroxisomes

(2) Golgi bodies

(3) Polysomes

(4) Endoplasmic reticulum

Ans. (2)

05. The QRS complex in a standard ECG represents :

(1) Depolarisation of auricles

(2) Depolarisation of ventricles

(3) Repolarisation of ventricles

(4) Repolarisation of auricles

Ans. (2)

06. Match the following columns and select the **correct** option.

	Column I		**Column II**
(a)	Floating ribs	(i)	Located between second and seventh ribs
(b)	Acromion	(ii)	Head of the Humerus
(c)	Scapula	(iii)	Clavicle
(d)	Glenoid cavity	(iv)	Do not connect with the sternum

	(a)	**(b)**	**(c)**	**(d)**
(1)	(i)	(iii)	(ii)	(iv)
(2)	(iii)	(ii)	(iv)	(i)
(3)	(iv)	(iii)	(i)	(ii)
(4)	(ii)	(iv)	(i)	(iii)

Ans. (3)

07. Experimental verification of the chromosomal theory of inheritance was done by :

(A) Sutton

(B) Boveri

(C) Morgan

(D) Mendel

Ans. (3)

08. Identify the **incorrecrt** statement.

(1) Sapwood is involved in conduction of water and minerals from root to leaf.

(2) Sapwood is the innermost secondary xylem and is lighter in colour.

(3) Due to deposition of tannins, resins, oils etc., heart wood is dark in colour.

(4) Heart wood does not conduct water but gives mechanical support

Ans. (2)

09. Match the following columns and select the **correct** option.

	Column I		**Column II**
(a)	Pituitary gland	(i)	Grave's disease
(b)	Thyroid gland	(ii)	Diabetes mellitus
(c)	Adrenal gland	(iii)	Diabetes insipidus
(d)	Pancreas	(iv)	Addison's disease

	(a)	**(b)**	**(c)**	**(d)**
(1)	(iii)	(ii)	(i)	(iv)
(2)	(iii)	(i)	(iv)	(ii)
(3)	(ii)	(i)	(iv)	(iii)
(4)	(iv)	(iii)	(i)	(ii)

Ans. (2)

10. Match the organism with its use in biotechnology.

(a)	*Bacillus thuringiensis*	(i)	Cloning vector
(b)	*Thermus aquaticus*	(ii)	Construction of first rDNA molecule
(c)	*Agrobacterium tumefaciens*	(iii)	DNA polymerase
(d)	*Salmonella typhimurium*	(iv)	Cry proteins

Select the correct option from the following:

	(a)	**(b)**	**(c)**	**(d)**
(1)	(iv)	(iii)	(i)	(ii)
(2)	(iii)	(ii)	(iv)	(i)
(3)	(iii)	(iv)	(i)	(ii)
(4)	(ii)	(iv)	(iii)	(i)

Ans. (1)

11. Identify the substances having glycosidic bond and peptide bond, respectively in their structure :

(1) Glycerol, trypsin

(2) Cellulose, lecithin

(3) Inulin, insulin

(4) Chitin, cholesterol

Ans. (3)

12. Name the enzyme that facilitates opening of DNA helix during transcription.

(1) DNA helicase

(2) DNA polymerase

(3) RNA polymerase

(4) DNA ligase

Ans. (3)

13. If the head of cockroach is removed, it may live for few days because :

(1) the cockroach does not have nervous system.

(2) the head holds a small proportion of a nervous system while the rest is situated along the ventral part of its body.

(3) the head holds a $1/3^{rd}$ of a nervous system while the rest is situated along the dorsal part of its body.

(4) the supra-oesophageal ganglia of the cockroach are situated in ventral part of abdomen.

Ans. (2)

14. Select the **correct** events that occur during inspiration.

(a) Contraction of diaphragm

(b) Contraction of external inter-costal muscles

(c) Pulmonary volume decreases

(d) Intra pulmonary pressure increases

(1) (c) and (d)

(2) (a), (b) and (d)

(3) only (d)

(4) (a) and (b)

Ans. (4)

15. By which method was a new breed 'Hisardale' of sheep formed by using Bikaneri ewes and Marino rams ?

(1) Mutational breeding

(2) Cross breeding

(3) Inbreeding

(4) Out crossing

Ans. (2)

16. Which one of the following is the most abundant protein in the animals ?

(1) Collagen

(2) Lectin

(3) Insulin

(4) Haemoglobin

Ans. (1)

17. How many true breeding pea plant varieties did Mendel select as pairs, which were similar except in one character with contrasting traits ?

(1) 2

(2) 14

(3) 8

(4) 4

Ans. (2)

18. The body of the ovule is fused within the funicle at :

(1) Micropyle

(2) Nucellus

(3) Chalaza

(4) Hilum

Ans. (4)

19. Which of the following is **correct** about viroids ?

(1) They have free RNA without protein coat.

(2) They have DNA with protein coat.

(3) They have free DNA without protein coat.

(4) They have RNA with protein coat.

Ans. (1)

20. The number of substrate level phosphorylations in one turn of citric acid cycle is :

(1) One

(2) Two

(3) Three

(4) Zero

Ans. (1)

21. The product(s) of reaction catalyzed by nitrogenase in root nodules of leguminous plants is/are :

(1) Nitrate alone

(2) Ammonia and oxygen

(3) Ammonia and hydrogen

(4) Ammonia alone

Ans. (3)

22. Match the following diseases with the causative organism and select the **correct** option.

	Column I		**Column II**
(a)	Typhoid	(i)	*Wuchereria*
(b)	Pneumonia	(ii)	*Plasmodium*
(c)	Filariasis	(iii)	*Salmonella*
(d)	Malaria	(iv)	*Haemophilus*

	(a)	**(b)**	**(c)**	**(d)**
(1)	(iii)	(iv)	(i)	(ii)
(2)	(ii)	(i)	(iii)	(iv)
(3)	(iv)	(i)	(ii)	(iii)
(4)	(i)	(iii)	(ii)	(iv)

Ans. (1)

23. Form his experiments, S.L. Miller produced amino acids by mixing the following in a closed flask :

(1) CH_3, H_2, NH_4 and water vapor at 800°C

(2) CH_4, H_2, NH_3 and water vapor at 600°C

(3) CH_3, H_2, NH_3 and water vapor at 600°C

(4) CH_4, H_2, NH_3 and water vapor at 800°C

Ans. (4)

24. Which of the following statements is **correct** ?

(1) Adenine pairs with thymine through one H-bond.

(2) Adenine pairs with thymine through three H-bonds.

(3) Adenine does not pair with thymine.

(4) Adenine pairs with thymine through two H-bonds.

Ans. (4)

25. Match the following with respect to meiosis :

	Column I		**Column II**
(a)	Zygotene	(i)	Terminalization
(b)	Pachytene	(ii)	Chiasmata
(c)	Diplotene	(iii)	Crosing over
(d)	Diakinesis	(iv)	Synapsis

Select the **correct** option from the following :

	(a)	**(b)**	**(c)**	**(d)**
(1)	(iv)	(iii)	(ii)	(i)
(2)	(i)	(ii)	(iv)	(iii)
(3)	(ii)	(iv)	(iii)	(i)
(4)	(iii)	(iv)	(i)	(ii)

Ans. (1)

26. Choose the **correct** pair from the following :

(1) Polymerases - Break the DNA into fragments

(2) Nucleases - Separate the two strands of DNA

(3) Exonucleases - Make cuts at specific positions within DNA

(4) Ligases - Join the two DNA molecules

Ans. (4)

27. Select the option including all sexually transmitted diseases.

(1) Gonorrhoea, Malaria, Genital herpes

(2) AIDS, Malaria, Filaria

(3) Cancer, AIDS, Syphilis

(4) Gonorrhoea, Syphilis, Genital herpes

Ans. (4)

28. Embryological support for evolution was disapproved by :

(1) Alfred Wallace

(2) Charles Darwin

(3) Oparin

(4) Karl Ernst von Baer

Ans. (4)

29. The roots that originate from the base of the stem are:

(1) Primary roots

(2) Prop roots

(3) Lateral roots

(4) Fibrous roots

Ans. (4)

30. In gel electrophoresis, separated DNA fragments can be visualized with the help of :

(1) Ethidium bromide in UV radiation

(2) Acetocarmine in UV radiation

(3) Ethidium bromide in infrared radiation

(4) Acetocarmine in bright blue light

Ans. (1)

31. Which of the following hormone levels will cause release of ovum (ovulation) from the graffian follicle?

(1) High concentration of Progesterone

(2) Low concentration of LH

(3) Low concentration of FSH

(4) High concentration of Estrogen

Ans. (4)

32. Goblet cells of alimentary canal are modified from :

(1) Columnar epithelial cells

(2) Chondrocytes

(3) Compound epithelial cells

(4) Squamous epithelial cells

Ans. (1)

33. Snow-blindness in Antarctic region is due to :

(1) Inflammation of cornea due to high dose of UV-B radiation

(2) High reflection of light from snow

(3) Damage to retina caused by infra-red rays

(4) Freezing of fluids in the eye by low temperature

Ans. (1)

34. Match the following concerning esential elements and their functions in plants :

(a)	Iron	(i)	Photolysis of water
(b)	Zinc	(ii)	Pollen germination
(c)	Boron	(iii)	Required for chlorophyll biosynthesis
(d)	Manganese	(iv)	IAA biosynthesis

Select the **correct** option :

	(a)	**(b)**	**(c)**	**(d)**
(1)	(iv)	(iii)	(ii)	(i)
(2)	(iii)	(iv)	(ii)	(i)
(3)	(iv)	(i)	(ii)	(iii)
(4)	(ii)	(i)	(iv)	(iii)

Ans. (2)

35. Bt cotton variety that was developed by the introduction of toxin gene of *Bacillus thuringiensis* (Bt) is resistant to :

(1) Fungal diseases

(2) Plant nematodes

(3) Insect predators

(4) Insect pests

Ans. (4)

36. Ray florets have :

(1) Superior ovary

(2) Hypogynous ovary

(3) Half inferior ovary

(4) Inferior ovary

Ans. (4)

37. Montreal protocol was signed in 1987 for control of :

(1) Emission of ozone depleting substances

(2) Release of Green House gases

(3) Disposal of e-wastes

(4) Transport of Genetically modified organisms from one country to another

Ans. (1)

38. Identify the **wrong** statement with regard to Restriction Enzymes.

(1) They cut the strand of DNA at palindromic sites.

(2) They are useful in genetic engineering.

(3) Sticky ends can be joined by using DNA ligases.

(4) Each restriction enzyme functions by inspecting the length of a DNA sequence.

Ans. (3)

39. The infectious stage of *Plasmodium* that enters the human body is :

(1) Sporozoites

(2) Female gametocytes

(3) Male gametocytes

(4) Trophozoites

Ans. (1)

40. Meiotic division of the secondary oocyte is completed:

(1) At the time of copulation

(2) After zygote formation

(3) At the time of fusion of a sperm with an ovum

(4) Prior to ovulation

Ans. (3)

41. The Oxygenation activity of RuBisCo enzyme in photorespiration leads to the formation of :

(1) 1 molecule of 3-C compound

(2) 1 molecule of 6-C compound

(3) 1 molecule of 4-C compound and 1 molecule of 2-C compound

(4) 2 molecules of 3-C compound

Ans. (1)

42. Which of the following statements are **true** for the phylum Chordata?

(a) In Urochordata notochord extends from head to tail and it is present throughout their life.

(b) In Vertebrata notochord is present during the embryonic period only.

(c) Central nervous system is dorsal and hollow.

(d) Chordata is divided into 3 subphyla: Hemichordata, Tunicata and Cepholochordata.

(1) (c) and (a)

(2) (a) and (b)

(3) (b) and (c)

(4) (d) and (c)

Ans. (3)

43. Match the following columns and select the **correct** option.

	Column I		**Column II**
(a)	*Clostridium butylicum*	(i)	Cyclosporin-A
(b)	*Trichoderma polysporum*	(ii)	Butyric Acid
(c)	*Monascus purpureus*	(iii)	Citric Acid
(d)	*Aspergillus niger*	(iv)	Blood cholesterol lowering agent

	(a)	**(b)**	**(c)**	**(d)**
(1)	(ii)	(i)	(iv)	(iii)
(2)	(i)	(ii)	(iv)	(iii)
(3)	(iv)	(iii)	(ii)	(i)
(4)	(iii)	(iv)	(ii)	(i)

Ans. (1)

44. Which of the following pairs is of unicellular algae ?

(1) *Gelidium* and *Gracilaria*

(2) *Anabaena* and *Volvox*

(3) *Chlorella* and *Spirulina*

(4) *Laminaria* and *Sargassum*

Ans. (3)

45. In light reaction, plastoquinone facilitates the transfer of electrons from :

(1) $Cytb_6f$ complex to PS-I

(2) PS-I to $NADP^+$

(3) PS-I to ATP synthase

(4) PS-II to $Cytb_6f$ complex

Ans. (4)

46. Presence of which of the following conditions in urine are indicative of Diabetes Mellitus ?

(1) Uremia and Renal Calculi

(2) Ketonuria and Glycosuria

(3) Renal calculi and Hyperglycaemia

(4) Uremia and Ketonuria

Ans. (2)

47. Secondary metabolites such as nicotine, strychnine and caffeine are produced by plants for their :

(1) Growth response

(2) Defence action

(3) Effect on reproduction

(4) Nutritive value

Ans. (2)

48. Which of the following would help in prevention of diuresis?

(1) Reabsorption of Na^+ and water from renal tubules due to aldosterone

(2) Atrial natriuretic factor causes vasoconstriction

(3) Decrease in secretion of renin by JG cells

(4) More water reabsorption due to undersectretion of ADH

Ans. (1)

49. Select the **Correct** match.

(1) Phenylketonuria - Autosomal dominant trait

(2) Sickle cell anaemia - Autosomal recessive trait, Chromosome - 11

(3) Thalassemia - X linked

(4) Haemophilia - Y linked

Ans. (2)

50. Which of the following is **not** an attribute of a population ?

(1) Natality

(2) Mortality

(3) Species interaction

(4) Sex ratio

Ans. (3)

51. Which of the following statements about inclusion bodies is **incorrect** ?

(1) These are involved in ingestion of food particles.

(2) They lie free in the cytoplasm.

(3) These represent reserve material in cytoplasm.

(4) They are not bound by any membrane.

Ans. (1)

52. The transverse section of a plant shows following anatomical features :

(a) Large number of scattered vascular bundles surrounded by bundle sheath.

(b) Large conspicuous parenchymatous ground tissue.

(c) Vascular bundles conjoint and closed.

(d) Pholem parenchyma absent.

Identify the category of plant and its part :

(1) Monocotyledonous root

(2) Dicotyledonous stem

(3) Dicotyledonous root

(4) Monocotyledonous stem

Ans. (4)

53. In relation to Gross primary productivity and Net primary productivity of an ecosystem, which one of the following statements is **correct** ?

(1) Gross primary productivity is always more than net primary productivity.

(2) Gross primary productivity and Net primary productivity are one and same.

(3) There is no relationship between Gross primary productivity and Net primary productivity.

(4) Gross primary productivity is always less than net primary productivity.

Ans. (1)

54. In water hyacinth and water lily, pollination takes place by:

(1) water currents only

(2) wind and water

(3) insects and water

(4) insects or wind

Ans. (4)

55. Which of the following is put into Anaerobic sludge digester for further sewage treatment ?

(1) Floating debris

(2) Effluents of primary treatment

(3) Activated sludge

(4) Primary sludge

Ans. (3)

56. The process responsible for facilitating loss of water in liquid form from the tip of grass blades at night and in early morning is :

(1) Root pressure

(2) Imbibition

(3) Plasmolysis

(4) Transpiration

Ans. (1)

57. Cubiodal epithelium with brush border of microvilli is found in :

(1) ducts of salvary glands

(2) proximal convoluted tubule of nephron

(3) eustachian tube

(4) lining of interstine

Ans. (2)

58. Select the **correct** statement

(1) Glucagon is associated with hypoglycemia.

(2) Insulin acts on pancreatic cells and adipocytes.

(3) Insulin is associated with hyperglycemia.

(4) Glucocorticoids stimulate gluconeogenesis.

Ans. (4)

59. Which of the following is **not** an inhibitory substance governing seed dormancy ?

(1) Abscisic acid

(2) Phenolic acid

(3) Para-ascorbic acid

(4) Gibberellic acid

Ans. (4)

60. According to Robert May, the global species diversity is about :

(1) 20 million

(2) 70 million

(3) 7 million

(4) 1.5 million

Ans. (3)

61. Match the trophic levels with their **correct** species examples in grassland ecosystem.

(a)	Fourth trophic level	(i)	Crow
(b)	Second trophic level	(ii)	Vulture
(c)	First trophic level	(iii)	Rabbit
(d)	Third trophic level	(iv)	Grass

	(a)	**(b)**	**(c)**	**(d)**
(1)	(iii)	(ii)	(i)	(iv)
(2)	(iv)	(iii)	(ii)	(i)
(3)	(i)	(ii)	(iii)	(iv)
(4)	(ii)	(iii)	(iv)	(i)

Ans. (4)

62. The first phase of translation is :

(1) Recognition of DNA molecule

(2) Aminoacylation of tRNA

(3) Recognition of an anti-codon

(4) Binding of mRNA to ribosome

Ans. (2)

63. Strobili or cones are found in :

(1) *Pteris*

(2) *Marchantia*

(3) *Equisetum*

(4) *Salvinia*

Ans. (3)

64. Match the following columns and select the **correct** option

	Column - I		**Column - II**
(a)	6-15 pairs of gill slits	(i)	*Trygon*
(b)	Heterocercal caudal fin	(ii)	Cyclostomes
(c)	Air Bladder	(iii)	Chondrichthyes
(d)	Poison sting	(iv)	Osteichthyes

	(a)	**(b)**	**(c)**	**(d)**
(1)	(iii)	(iv)	(i)	(ii)
(2)	(iv)	(ii)	(iii)	(i)
(3)	(i)	(iv)	(iii)	(ii)
(4)	(ii)	(iii)	(iv)	(i)

Ans. (4)

65. Some dividing cells exit the cell cycle and enter vegetative inactive stage. This is called quiescent stage (G_0). This process occurs at the end of :

(1) G_1 phase

(2) S phase

(3) G_2 phase

(4) M phase

Ans. (1)

66. Identify the **correct** statement with reference to human digestive system.

(1) Serosa is the innermost layer of the alimentary canal.

(2) Ileum is a highly coiled part.

(3) Vermiform appendix arises from duodenum.

(4) Ileum opens into small intestine.

Ans. (2)

67. In which of the following techniques, the embryos are transferred to assist those females who cannot conceive ?

(1) GIFT and ZIFT

(2) ICSI and ZIFT

(3) GIFT and ICSI

(4) ZIFT and IUT

Ans. (4)

68. The plant parts which consist of two generations - one within the other :

(a) Pollen grains inside the anther

(b) Germinated pollen grain with two male gametes

(c) Seed inside the fruit

(d) Embryo sac inside the ovule

(1) (a), (b) and (c)

(2) (c) and (d)

(3) (a) and (d)

(4) (a) only

Ans. (3)

69. Dissolution of the synaptonemal complex occurs during :

(1) Zygotene

(2) Diplotene

(3) Leptotene

(4) Pachytene

Ans. (2)

70. Match the following columns and select the **correct** option.

	Column - I		**Column - II**
(a)	Organ of Corti	(i)	Connects middle ear and pharynx
(b)	Cochlea	(ii)	Coiled part of the labyrinth
(c)	Eustachian tube	(iii)	Attached to the oval window
(d)	Stapes	(iv)	Located on the basilar membrane

	(a)	**(b)**	**(c)**	**(d)**
(1)	(iii)	(i)	(iv)	(ii)
(2)	(iv)	(ii)	(i)	(iii)
(3)	(i)	(ii)	(iv)	(iii)
(4)	(ii)	(iii)	(i)	(iv)

Ans. (2)

71. The ovary is half inferior in :

(1) Mustard

(2) Sunflower

(3) Plum

(4) Brinjal

Ans. (3)

72. Identify the basic amino acid from the following.

(1) Glutamic Acid

(2) Lysine

(3) Valine

(4) Tyrosine

Ans. (2)

73. Match the following columns and select the **correct** option.

	Column - I		Column - II
(a)	Eosinophils	(i)	Immune response
(b)	Basophils	(ii)	Phagocytosis
(c)	Neutrophils	(iii)	Release histaminase, destructive enzymes
(d)	Lymphocytes	(iv)	Release granules containing histamine

	(a)	**(b)**	**(c)**	**(d)**
(1)	(iv)	(i)	(ii)	(iii)
(2)	(i)	(ii)	(iv)	(iii)
(3)	(ii)	(i)	(iii)	(iv)
(4)	(iii)	(iv)	(ii)	(i)

Ans. (4)

74. Match the following :

	Column - I		**Column - II**
(a)	Inhibitor of catalytic activity	(i)	Ricin
(b)	Possess peptide bonds	(ii)	Malonate
(c)	Cell wall material in fungi	(iii)	Chitin
(d)	Secondary metabolite	(iv)	Collagen

Choose the **correct** option from the following :

	(a)	**(b)**	**(c)**	**(d)**
(1)	(iii)	(i)	(iv)	(ii)
(2)	(iii)	(iv)	(i)	(ii)
(3)	(ii)	(iii)	(i)	(iv)
(4)	(ii)	(iv)	(iii)	(i)

Ans. (4)

75. Identify the correct statement with regard to G_1 phase (Gap 1) of interphase.

(1) Reorganisation of all cell components takes place.

(2) Cell is metabolically active, grows but does not replicate its DNA

(3) Nuclear Division takes place.

(4) DNA synthesis or replication takes place.

Ans. (2)

76. Name the plant growth regulator which upon spraying on sugarcane crop, increases the length of stem, thus increasing the yield of sugarcane crop.

(1) Gibberellin

(2) Ethylene

(3) Abscisic acid

(4) Cytokinin

Ans. (1)

77. Identify the **wrong** statement with reference to the gene 'I' that controls ABO blood groups.

(1) A person will have only two of the three alleles.

(2) When I^A and I^B are present together, they express same type of sugar.

(3) Allele 'i' does not produce any sugar.

(4) The gene (I) has three alleles.

Ans. (2)

78. Identify the **wrong** statement with reference to immunity.

(1) When ready-made antibodies are directly given, it is called "Passive immunity".

(2) Active immunity is quick and gives full response.

(3) Foetus receives some antibodies from mother, it is an example for passive immunity.

(4) When exposed to antigen (living or dead) antibodies are produced in the host's body. It is called "Active immunity".

Ans. (2)

79. The enzyme enterokinase helps in conversion of :

(1) trypsinogen into trypsin

(2) caseinogen into casein

(3) pepsinogen into pepsin

(4) protein into polypeptides

Ans. (1)

80. The specific palindromic sequence which is recognized by EcoRI is :

(1) 5'-GGAACC-3'
3'-CCTTGG-5'

(2) 5'-CTTAAG-3'
3'-GAATTC-5'

(3) 5'-GGATCC-3'
3'-CCTAGG-5'

(4) 5'-GAATTC-3'
3'-CTTAAG-5'

Ans. (4)

81. Match the following columns and select the **correct** option.

	Column - I		**Column - II**
(a)	Bt cotton	(i)	Gene therapy
(b)	Adenosine deaminese deficiency	(ii)	Cellular defence
(c)	RNAi	(iii)	Detection of HIV infection
(d)	PCR	(iv)	*Bacillus thuringiensis*

	(a)	**(b)**	**(c)**	**(d)**
(1)	(iii)	(ii)	(i)	(iv)
(2)	(ii)	(iii)	(iv)	(i)
(3)	(i)	(ii)	(iii)	(iv)
(4)	(iv)	(i)	(ii)	(iii)

Ans. (4)

82. Floridean starch has structure similar to :

(1) Amylopectin and glycogen

(2) Mannitol and algin

(3) Laminarin and cellulose

(4) Starch and cellulose

Ans. (1)

83. Which of the following statements is **not correct** ?

(1) The proinsulin has an extra peptide called C-peptide.

(2) The functional insulin has A and B chains link together by hydrogen bonds.

(3) Genetically engineered insulin is produced in *E-Coli*.

(4) In man insulin is synthesised as a proinsulin.

Ans. (2)

84. Flippers of Penguins and Dolphins are examples of :

(1) Convergent evolution

(2) Industrial melanism

(3) Natural selection

(4) Adaptive radiation

Ans. (1)

85. Which of the following refer to **correct** example(s) of organisms which have evolved due to changes in environment brought about by anthropogentic action ?

(a) Darwin's Finches of Galapagos islands.

(b) Herbicide resistant weeds.

(c) Drug resistant eukaryotes.

(d) Man-created breeds of domesticated animals like dogs.

(1) (a) and (c)

(2) (b), (c) and (d)

(3) only (d)

(4) only (a)

Ans. (2)

86. Identify the **wrong** statement with reference to transport of oxygen.

(1) Partial pressure of CO_2 can interfere with O_2 binding with haemoglobin.

(2) Higher H^+ conc. in alveoli favours the formation of oxyhaemoglobin.

(3) Low pCO_2 in alveoli favours the formation of oxyhaemoglobin.

(4) Binding of oxygen with haemoglobin is mainly related to partial pressure of O_2.

Ans. (2)

87. The process of growth is maximum during :

(1) Lag phase

(2) Senescence

(3) Dormancy

(4) Log phase

Ans. (4)

88. Which of the following regions of the globe exhibits highest species diversity ?

(1) Madagascar

(2) Himalayas

(3) Amazon forests

(4) Western Ghats of India

Ans. (3)

89. The sequence that controls the copy number of the linked DNA in the vector, is termed :

(1) Ori site

(2) Palindromic sequence

(3) Recognition site

(4) Selectable marker

Ans. (1)

90. Match the following columns and select the **correct** option.

	Column I		**Column II**
(a)	Placenta	(i)	Androgens
(b)	Zona pellucida	(ii)	Human Chorionic Gonadotropic (hCG)
(c)	Bulbo-urethral glands	(iii)	Layer of the ovum
(d)	Leydig cells	(iv)	Lubrication of the Penis

	(a)	**(b)**	**(c)**	**(d)**
(1)	(i)	(iv)	(ii)	(iii)
(2)	(iii)	(ii)	(iv)	(i)
(3)	(ii)	(iii)	(iv)	(i)
(4)	(iv)	(iii)	(i)	(ii)

Ans. (3)

CHEMISTRY

91. Sucrose on hydrolysis gives :

(1) α -D-Glucose + β -D-Glucose

(2) α -D-Glucose + β -D-Fructose

(3) α -D-Fructose + β -D-Fructose

(4) β -D-Glucose + α -D-Fructose

Ans. (2)

92. Elimination reaction of 2-Bromo-pentane to form pen-2-ene is :

(a) β -Elimination reaction

(b) Follows Zaitsev rule

(c) Dehydrohalogenation reaction

(d) Dehydration reaction

(1) (a), (c), (d)

(2) (b), (c), (d)

(3) (a), (b), (d)

(4) (a), (b), (c)

Ans. (4)

93. The number of Faradays (F) required to produce 20g of calcium from molten $CaCl_2$ (Atomic mass of Ca = $40\,g\,mol^{-1}$) is :

(1) 2

(2) 3

(3) 4

(4) 1

Ans. (1)

94. An element has a body centered cubic (bcc) structure with a cell edge of 288pm. The atomic radius is :

(1) $\frac{\sqrt{2}}{4}\times 288\,pm$

(2) $\frac{}{\sqrt{3}}\times 288\,pm$

(3) $\frac{}{\sqrt{2}}\times 288\,pm$

(4) $\frac{\sqrt{3}}{4}\times 288\,pm$

Ans. (4)

95. HCl was passed through a solution of $CaCl_2$, $MgCl_2$ and NaCl. Which of the following compound(s) crystallise(s) ?

(1) Only NaCl

(2) Only $MgCl_2$

(3) NaCl, $MgCl_2$ and $CaCl_2$

(4) Both $MgCl_2$ and $CaCl_2$

Ans. (1)

96. Find out the solubility of $Ni(OH)_2$ in 0.1M NaOH. Given that the ionic product of $Ni(OH)_2$ is 2×10^{-15}

(1) $2\times10^{-8}M$

(2) $1\times10^{-13}M$

(3) $1\times10^{8}M$

(4) $2\times10^{-13}M$

Ans. (4)

97. For the reaction, $2Cl(g)\rightarrow Cl_2(g)$, the correct option is :

(1) $\Delta_rH>0$ and $\Delta_rS<0$

(2) $\Delta_rH<0$ and $\Delta_rS>0$

(3) $\Delta_rH<0$ and $\Delta_rS<0$

(4) $\Delta_rH>0$ and $\Delta_rS>0$

Ans. (3)

98. Which of the following is the correct order of increasing field strength of ligands to form coordination compounds ?

(1) $SCN^-<F^-<CN^-<C_2O_4^{2-}$

(2) $F^-<SCN^-<C_2O_4^{2-}<CN^-$

(3) $CN^-<C_2O_4^{2-}<SCN^-<F^-$

(4) $SCN^-<F^-<C_2O_4^{2-}<CN^-$

Ans. (4)

99. The calculated spin only magnetic moment of Cr^{2+} ion is :

(1) 4.90 BM

(2) 5.92 BM

(3) 2.84 BM

(4) 3.87 BM

Ans. (1)

100. Which of the following set of molecules will have zero dipole moment ?

(1) Boron trifluoride, hydrogen fluoride, carbondioxide, 1, 3-dichlorobenzene

(2) Nitrogen trifluoride, beryllium difluoride, water, 1, 3-dichlorobenzene

(3) Boron trifluoride, beryllium difluoride, carbon dioxide, 1, 4-dichlorobenzene

(4) Ammonia, beryllium difluoride, water, 1, 4-dichlorobenzene

Ans. (3)

101. The following metal ion activates many enzymes, participates in the oxidation of glucose to produce ATP and with Na, is responsible for the transmission of nerve signals

(1) Copper

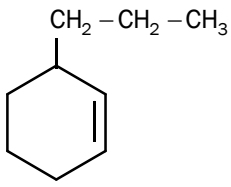
(2) Calcium

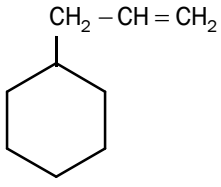
(3) Potassium

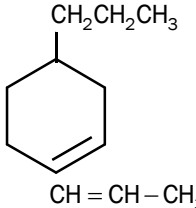
(4) Iron

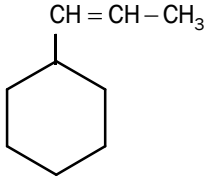
Ans. (3)

102. Al alkene on ozonolysis gives methanal as one of the product. Its structure is

(1) 

(2) 

(3) 

(4) 

Ans. (2)

103. The rate constant for a first order reaction is $4.606 \times 10^{-3} s^{-1}$. The time required to reduce 2.0 g of the reactant to 0.2 g is

(1) 200 s

(2) 500 s

(3) 1000 s

(4) 100 s

Ans. (2)

104. Reaction between acetone and methylmagnesium chloride followed by hydrolysis will give :

(1) Sec. butyl alcohol

(2) Tert. butyl alcohol

(3) Isobutyl alcohol

(4) Isopropyl alcohol

Ans. (2)

105. Which of the following is a natural polymer ?

(1) poly (Butadiene - styrene)

(2) polybutadiene

(3) poly (Butadiene - acrylonitrile)

(4) cis - 1, 4 - polyisoprene

Ans. (4)

106. Identify the correct statements from the following :

(a) $CO_2(g)$ is used as refrigerant for ice-cream and frozen food

(b) The structure of C_{60} contains twelve six carbon rings and twenty fice carbon rings

(c) ZSM - 5, a type of zeolite, is used to convert alcohols into gasoline

(d) CO is colourless and odourless gas

(1) (a) and (c) only

(2) (b) and (c) only

(3) (c) and (d) only

(4) (a), (b) and (c) only

Ans. (3)

107. The correct option for free expansion of an ideal gas under adiabatic condition is :

(1) $q = 0, \Delta T = 0$ and $w > 0$

(2) $q < 0, \Delta T = 0$ and $w = 0$

(3) $q > 0, \Delta T > 0$ and $w > 0$

(4) $q = 0, \Delta T = 0$ and $w = 0$

Ans. (4)

108. Which of the following oxoacid of sulphur has O - O - linkage?

(1) H_2SO_4 , sulphuric acid

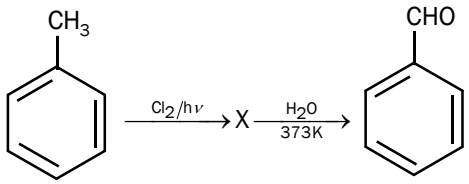
(2) $H_2S_2O_8$, peroxidisulphuric acid

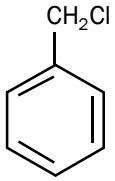
(3) $H_2S_2O_7$, pyrosulphuric acid

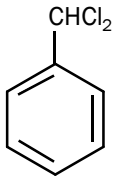
(4) H_2SO_3 , sulphurous acid

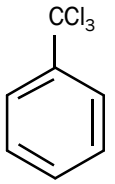
Ans. (2)

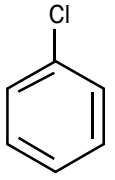
109. Identify compound X in the following sequence of reactions:



(1) 

(2) 

(3) 

(4) 

Ans. (2)

110. The number of protons, neutrons and electrons in $^{175}_{71}Lu$, respectively, are:

(1) 104, 71 and 71

(2) 71, 71 and 104

(3) 175, 104 and 71

(4) 71, 104 and 71

Ans. (4)

111. Identify the incorrect statement

(1) The transition metals and their compounds are known for their catalytic activity due to their ability to adopt multiple oxidation states and to form complexes

(2) Interstitial compounds are those that are formed when small atoms like H, C or N are trapped inside the crystal lattice of metals

(3) The oxidation states of chromium in CrO_4^{2-} and $Cr_2O_7^{2-}$ are not the same

(4) $Cr^{2+}(d^4)$ is a stronger reducing agent than $Fe^{2+}(d^6)$ in water.

Ans. (3)

112. Which of the following is a cationic detergent?

(1) Sodium sterate

(2) Cetyltrimethyl ammonium bromide

(3) Sodium dodecylbenzene sulphonate

(4) Sodium lauryl sulphate

Ans. (2)

113. The freezing point depression constant (K_f) of benzene is 5.12 K $kg\,mol^{-1}$. The freezing point depression for the solution of molality 0.078 m containing a non-electrolyte solute in benzene is (rounded off upto two decimal places):

(1) 0.80 K

(2) 0.40 K

(3) 0.60 K

(4) 0.20 K

Ans. (2)

114. Identify the incorrect match

	Name		**IUPAC Official Name**
(a)	Unnilunium	(i)	Mendelevium
(b)	Unniltrium	(ii)	Lawrencium
(c)	Unnilhexium	(iii)	Seaborgium
(d)	Unununnium	(iv)	Darmastadtium

(1) (b), (ii)

(2) (c), (iii)

(3) (d), (iv)

(4) (a), (i)

Ans. (3)

115. The mixture which shows positive deviation from Raoult's law is:

(1) Benzene + Toulene

(2) Acetone + Chloroform

(3) Chloroethane + Bromoethane

(4) Ethanol + Acetone

Ans. (4)

116. Match the following:

	Oxide		Nature
(a)	CO	(i)	Basic
(b)	BaO	(ii)	Neutral
(c)	Al_2O_3	(iii)	Acidic
(d)	Cl_2O_7	(iv)	Amphoteric

Which of the following is correct option?

	(a)	(b)	(c)	(d)
(1)	(ii)	(i)	(iv)	(iii)
(2)	(iii)	(iv)	(i)	(ii)
(3)	(iv)	(iii)	(ii)	(i)
(4)	(i)	(ii)	(iii)	(iv)

Ans. (1)

117. Which one of the following has maximum number of atoms?

(1) 1 g of Mg(s) [Atomic mass of Mg = 24]

(2) 1g of O_2 (g) [Atomic mass of O = 16]

(3) 1 g of Li(s) [Atomic mass of Li = 7]

(4) 1 g of Ag(s) [Atomic mass of Ag = 108]

Ans. (3)

118. Reaction between benzaldehyde and acetophenone in presence of dilute NaOH is known as:

(1) Cannizzaro's reaction

(2) Cross Cannizzaro's rection

(3) Cross Aldol condensation

(4) Aldol condensation

Ans. (3)

119. A tertiary butyl carbocation is most stable than a secondary butyl carbocation because of which of the following?

(1) + Reflect of - CH_3 groups

(2) - Reflect of - CH_3 groups

(3) Hyperconjugation

(4) - I effect of - CH_3 groups

Ans. (3)

120. Which of the following is not correct about carbon monboxide?

(1) It reduces oxygen carrying ability of blood

(2) The carboxyhaemoglobin (haemoglobin bound to CO) is less stable than oxyhaemoglobin.

(3) It is produced due to incomplete combustion

(4) It forms carboxyhaemoglobin

Ans. (2)

121. Which of the following is a basic amino acid?

(1) Alanine (2) Tyrosine

(3) Lysine (4) Serine

Ans. (3)

122. Urea reacts with water to form A which will decompose to form B. B when passed through Cu^{2+} (aq), deep blue colour solution C is formed. What is formula of C from the following?

(1) $[Cu(NH_3)_4]^{2+}$

(2) $Cu(OH)_2$

(3) $CuCO_3.Cu(OH)_2$

(4) $CuSO_4$

Ans. (1)

123. A mixture of N_2 and Ar gases in a cylinder contains 7g of N_2 and 8 g of Ar. If the total pressure of the mixture of the gases in the cylinder is 27 bar, the partial pressure of N_2 is

[Use of atomic masses (in $g\,mol^{-1}$): N = 14, Ar = 40]

(1) 12 bar

(2) 15 bar

(3) 18 bar

(4) 9 bar

Ans. (2)

124. Identify the correct statement from the following:

(1) Blister copper has blistered appearance due to evolution of CO_2

(2) Vapour phase refining is carried out for Nickle by Van Arkel method

(3) Pig iron can be moulded into a variety of shapes

(4) Wrought iron is impure iron with 4% carbon

Ans. (3)

125. Hydrolysis of sucrose is given by the following reaction

$$\text{Sucrose} + H_2O \rightleftharpoons \text{Glucose} + \text{Fructose}$$

If the equilibrium constant (K_c) is 2×10^{13} at 300 K, the value $\Delta_r G^{\ominus}$ of at the same temperature will be

(1) $8.314\ \text{J mol}^{-1}\text{K}^{-1}\times300\ \text{K}\times\ln(2\times10^{13})$

(2) $8.314\ \text{J mol}^{-1}\text{K}^{-1}\times300\ \text{K}\times\ln(3\times10^{13})$

(3) $-8.314\ \text{J mol}^{-1}\text{K}^{-1}\times300\ \text{K}\times\ln(4\times10^{13})$

(4) $-8.314\ \text{J mol}^{-1}\text{K}^{-1}\times300\ \text{K}\times\ln(2\times10^{13})$

Ans. (4)

126. Identify a molecule which does not exist

(1) Li_2

(2) C_2

(3) O_2

(4) He_2

Ans. (4)

127. An increase in the concentration of the reactants of a reaction leads to change in :

(1) heat of reaction

(2) threshold energy

(3) collision frequency

(4) activation energy

Ans. (3)

128. Which of the followig alkane cannot be made in good yield by Wurtz reaction ?

(1) 2, 3 - Dimethylbutane

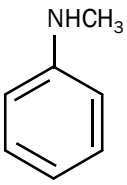
(2) n - Heptane

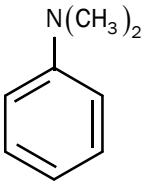
(3) n - Butane

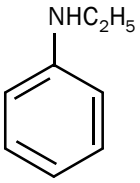
(4) n - Hexane

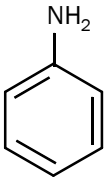
Ans. (2)

129. Which of the following amine will give teh carbylamine test ?

(1) 

(2) 

(3) 

(4) 

Ans. (4)

130. On electrolysis of dil. sulphuric acid using Platinum (Pt) electrode, the product obtained at anode will be :

(1) Oxygen gas

(2) H_2S gas

(3) SO_2 gas

(4) Hydrogen gas

Ans. (1)

131 What is the change in oxidation number of carbon in the following reaction ?

$$CH_4(g) + 4Cl_2(g) \rightarrow CCl_4(\ell) + 4HCl(g)$$

(1) 0 to +4

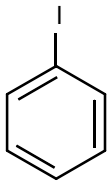
(2) -4 to +4

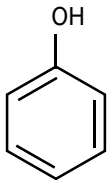
(3) 0 to -4

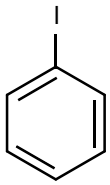
(4) +4 to +4

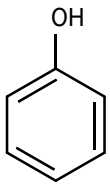
Ans. (2)

132. Anisole on cleavage with HI gives :

(1)  + CH_3OH

(2)  + C_2H_5I

(3)  + C_2H_5OH

(4)  + CH_3I

Ans. (4)

133. Measuring Zeta potential is useful in determining which property of colloidal solution ?

(1) Solubility

(2) Stability of the colloidal particles

(3) Size of the colloidal particles

(4) Viscosity

Ans. (2)

134. Paper chromatography is an example of :

(1) Partition chromatography

(2) Thin layer chromatography

(3) Column chromatography

(4) Adsorption chromatography

Ans. (1)

135. Match the following and identify the correct option.

(a)	$CO(g)+H_2(g)$	(i)	$Mg(HCO_3)_2+Ca(HCO_3)_2$
(b)	Temporary hardness of water	(ii)	An electron deficient hydride
(c)	B_2H_6	(iii)	Synthesis gas
(d)	H_2O_2	(iv)	Non-planar

	(a)	(b)	(c)	(d)
(1)	(iii)	(ii)	(i)	(iv)
(2)	(iii)	(iv)	(ii)	(i)
(3)	(i)	(iii)	(ii)	(iv)
(4)	(iii)	(i)	(ii)	(iv)

Ans. (4)

PHYSICS

136. A series LCR circuit is connected to an ac voltage source. When L is removed from the circuit, the phase difference between current and voltage is $\frac{\pi}{3}$. If instead C is removed from the circuit, the phase difference is again $\frac{\pi}{3}$ between current and voltage. The power factor of the circuit is :

(1) 0.5

(2) 1.0

(3) -1.0

(4) zero

Ans. (2)

137. A wire of length L, area of cross section A is hanging from a fixed support. The length of the wire changes to L_1 when mass M is suspended from its free end. The expression for Young's modulus is

(1) $\frac{Mg(L_1 - L)}{AL}$

(2) $\frac{MgL}{AL_1}$

(3) $\frac{MgL}{A(L_1 - L)}$

(4) $\frac{MgL_1}{AL}$

Ans. (3)

138. A long solenoid of 50 cm length having100 turns carries a current of 2.5 A. The magnetic field at the centre of the solenoid is:

$(\mu_0 = 4\pi \times 10^{-7}\ TmA^{-1})$

(1) 3.14×10^{-4} T

(2) 6.28×10^{-5} T

(3) 3.14×10^{-5} T

(4) 6.28×10^{-4} T

Ans. (4)

139. A ray is incident at an angle of incidence *i* on one surface of a small angle prism (with angle of prism A) and emerges normally from the opposite surface. If the refractive index of the material of the prism is μ, then the angle of incidence is nearly equal to :

(1) $\frac{2A}{\mu}$

(2) μA

(3) $\frac{\mu A}{2}$

(4) $\frac{A}{2\mu}$

Ans. (2)

140. In a certain region of space with volume $0.2m^3$, the electric potential is found to be 5V throughout. The magnitude of electric field in this region is:

(1) 0.5 N/C (2) 1 N/C

(3) 5 N/C (4) zero

Ans. (4)

141. For which one of the following, Bohr model is not valid?

(1) Singly ionised helium atom (He^+)

(2) Deuteron atom

(3) Singly ionised neon atom (Ne^+)

(4) Hydrogen atom

Ans. (3)

142. Light with an average flux of 20 W/cm^2 falls on a non-reflecting surface at normal incidence having surface area 20 cm^2. The energy received by the surface during time span of 1 minute is:

(1) 12×10^3 J

(2) 24×10^3 J

(3) 48×10^3 J

(4) 10×10^3 J

Ans. (2)

143. An electron is accelerated from rest through a potential didifference of V volt. If the de Broglie wavelength of the electron is 1.227×10^{-2} nm, the potential difference is :

(1) 10^2 V

(2) 10^3 V

(3) 10^4 V

(4) 10 V

Ans. (3)

144. A body weighs 72 N on the surface of the earth. What is the gravitational force on it, at a height equal to half the radius of the earth ?

(1) 32 N

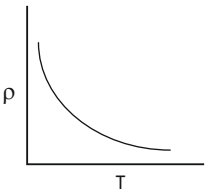
(2) 30 N

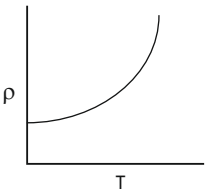
(3) 24 N

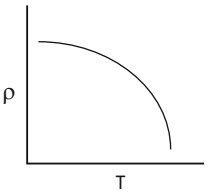
(4) 48 N

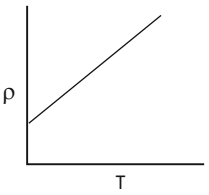
Ans. (1)

145. Which of the following graph represents the variation of resistivity (ρ) with temperature (T) for copper?

(1) 

(2) 

(3) 

(4) 

Ans. (2)

146. Light of frequency 1.5 times the threeshold frequency is incident on a photosensitive material. What will be the photoelectric current if the frequency is halved and intensity is doubled ?

(1) four times

(2) one-fourth

(3) zero

(4) double

Ans. (3)

147. A $40\mu F$ capacitor is connected to a 200 V , 50 Hz ac supply. The rms value of the current in the circuit is , nearly :

(1) 2.05 A

(2) 2.5 A

(3) 25.1 A

(4) 1.7 A

Ans. (2)

148. Assume that light of wavelenght 600 nm is coming from a star. The limit of resolution of telescope whose objective has a diameter of 2 m is :

(1) 1.83×10^{-7} rad

(2) 7.32×10^{-7} rad

(3) 6.00×10^{-7} rad

(4) 3.66×10^{-7} rad

Ans. (4)

149. A ball is thrown vertically downward with a velocity of 20 m/s from the top of a tower. It hits the ground after some time with a velocity of 80 m/s. The height of the tower is :(g=10 m/s^2)

(1) 340 m

(2) 320 m

(3) 300 m

(4) 360 m

Ans. (3)

150. A cylinder contains hydrogen gas at pressure of 249 kPa and temperature 27^0C.
Its density is : (R=8.3$Jmol^{-1}K^{-1}$)

(1) $0.2kg/m^3$

(2) $0.1kg/m^3$

(3) $0.02kg/m^3$

(4) $0.5kg/m^3$

Ans. (1)

151. When a uranium isotope $^{235}_{92}U$ is bombarded with a neutron, it generate $^{89}_{36}Kr$, three neutrons and :

(1) $^{91}_{40}Zr$

(2) $^{101}_{36}Kr$

(3) $^{103}_{36}Kr$

(4) $^{144}_{56}Ba$

Ans. (4)

152. The increase in the width of the depletion region in a p-n junction diode is due to :

(1) reverse bias only

(2) both forward bias and reverse bias

(3) increase in forward current

(4) forward bias only

Ans. (1)

153. The phase difference between displacement and acceleration of a particle in a simple harmonic motion is :

(1) $\frac{3\pi}{2}$rad

(2) $\frac{\pi}{2}$rad

(3) zero

(4) πrad

Ans. (4)

154. An iron rod of susceptibility 599 is subjected to a magnetising field of 1200 A m^{-1} . The permeability of the material of the rod is

$\left(\mu_0 = 4\pi\times10^{-7}\,TmA^{-1}\right)$

(1) $8.0\times10^{-5}\,TmA^{-1}$

(2) $2.4\pi\times10^{-5}\,TmA^{-1}$

(3) $2.4\pi\times10^{-7}\,TmA^{-1}$

(4) $2.4\pi\times10^{-4}\,TmA^{-1}$

Ans. (4)

155. The quantities of heat required to raise the temperature of two solid copper spheres of radii r_1 and r_2 $(r_1 = 1.5r_2)$ through 1 K are in the ratio:

(1) $\frac{9}{4}$

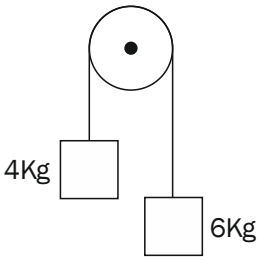
(2) $\frac{3}{2}$

(3) $\frac{5}{3}$

(4) $\frac{27}{8}$

Ans. (4)

156. Two bodies of mass 4 kg and 6kg are tied to the ends of a maasless string. The string passes over a pulley which is frictionless (see figure). The accleration of the system in terms of acceleration due to gravity (g) is :



(1) g/2

(2) g/5

(3) g/10

(4) g

Ans. (2)

157. The mean free path for a gas, with molecular diamter d and number density n can be expressed as :

(1) $\frac{1}{\sqrt{2}n\pi d^2}$

(2) $\frac{1}{\sqrt{2}n^2\pi d^2}$

(3) $\frac{1}{\sqrt{2}n^2\pi^2 d^2}$

(4) $\frac{1}{\sqrt{2}n\pi d}$

Ans. (1)

158. A short electric dipole has a dipole moment of 16×10^{-9} Cm . The electric potential due to the dipole at a point at a distance of 0.6 m from the centre of the dipole, situated on a line making an angle of 60^0 with the dipole axis is :

$\left(\frac{1}{4\pi\varepsilon_0} = 9\times10^9 Nm^2/C^2\right)$

(1) 200 V

(2) 400 V

(3) zero

(4) 50 V

Ans. (1)

159. Dimesions of stress are :

(1) $[ML^2T^{-2}]$

(2) $[ML^0T^{-2}]$

(3) $[ML^{-1}T^{-2}]$

(4) $[MLT^{-2}]$

Ans. (3)

160. The energy required to break one bond in DNA is 10^{-20} J. This value in eV is nearly :

(1) 0.6

(2) 0.06

(3) 0.006

(4) 6

Ans. (2)

161. In a guitar, two strings A and B made of same material are slightly out of tune and produce beats of frequency 6 Hz. when tension in B is slightly decreased, the beat frequency increases to 7 Hz. If the frequency of A is 530 Hz, the original frequency of B will be :

(1) 524 Hz

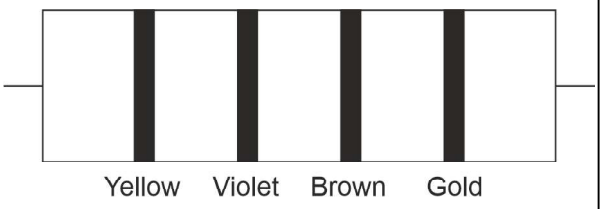
(2) 536 Hz

(3) 537 Hz

(4) 523 Hz

Ans. (1)

162. The color code of a resistance is given below :



The values of resistance and tolerance, respectively, are :

(1) $47k\Omega, 10\%$

(2) $4.7k\Omega, 5\%$

(3) $470\Omega, 5\%$

(4) $470k\Omega, 5\%$

Ans. (3)

163. The Brewsters angle i_b for an interface should be

(1) $30^0 < i_b < 45^0$

(2) $45^0 < i_b < 90^0$

(3) $i_b = 90^0$

(4) $0^0 < i_b < 30^0$

Ans. (2)

164. The capacitacne a parallel plate capacitor with air as medium is $6\mu F$. With the introduction of a dielectric medium, the capacitance becomes $30\mu F$. The permittivity of the medium is

$\left(\varepsilon_0 = 8.55 \times 10^{-12} C^2 N^{-1} m^{-2}\right)$

(1) $1.77 \times 10^{-12} C^2 N^{-1} m^{-2}$

(2) $0.44 \times 10^{-10} C^2 N^{-1} m^{-2}$

(3) $5.00 C^2 N^{-1} m^{-2}$

(4) $0.44 \times 10^{-13} C^2 N^{-1} m^{-2}$

Ans. (2)

165. Find the torque about the origin when a force of $3\hat{j}\,N$ acts on a particle whose position vector is $2\hat{k}\,m$.

(1) $6\hat{j}\,Nm$

(2) $-6\hat{i}\,Nm$

(3) $6\hat{k}\,Nm$

(4) $6\hat{i}\,Nm$

Ans. (2)

166. A resistance wire connected in the left gap of a metre bridge balances a 10Ω resistance in the right gap at a point which divides the bridge wire in the ratio 3 : 2. If the length of the resistance wire is 1.5 m, then the length of 1Ω of the resistance wire is :

(1) $1.0 \times 10^{-1} m$

(2) $1.5 \times 10^{-1} m$

(3) $1.5 \times 10^{-2} m$

(4) $1.0 \times 10^{-2} m$

Ans. (1)

167. For transistor action, which of the following statements is correct?

(1) Base, emitter and collector regions should have same size

(2) Both emitter junction as well as the colector junctions are forward biased

(3) The base region must be very thin and lightly doped

(4) Base, emitter and collector regions should have same doping concentrations

Ans. (3)

168. The ratio of contributions made by the electric field and magnetic field components to the intensity of an electromagnetic waves is :

(c= speed of electromagnetic waves)

(1) 1 : 1

(2) 1 : c

(3) $1 : c^2$

(4) c : 1

Ans. (1)

169. A charged particle having drift velocity of $7.5 \times 10^{-4} ms^{-1}$ in an electric field of $3 \times 10^{-10} Vm^{-1}$, has a mobility in $m^2 V^{-1} s^{-1}$ of

(1) 2.5×10^6

(2) 2.5×10^{-6}

(3) 2.25×10^{-15}

(4) 2.25×10^{15}

Ans. (1)

170. A spherical conductor of radius 10cm has a charge of $3.2\times10^{-7}C$ distributed uniformly. What is the magnitude of electric field at a point 15 cm from the centre of the sphere

$\left(\frac{1}{4\pi\epsilon_0}=9\times10^9 Nm^2/C^2\right)$

(1) $1.28\times10^5 N/C$

(2) $1.28\times10^6 N/C$

(3) $1.28\times10^7 N/C$

(4) $1.28\times10^4 N/C$

Ans. (1)

171. Taking into account of the significant figures, what is the value of 9.99 m - 0.0099 m?

(1) 9.98m

(2) 9.980 m

(3) 9.9 m

(4) 9.9801 m

Ans. (1)

172. In Young's double slit experiment, if the separation between coherent sources is halved and the distance of the screen from the coherent sources is doubled, then the fringe width becomes

(1) half

(2) four times

(3) one-fourth

(4) double

Ans. (2)

173. Two cyclinder A and B of equal capacity are connected to each other via a stop cock. A contains an ideal gas at standard temperature and pressure. B is completely evacuated. The entire system is thermally insulated. The stop cock is suddenly opened. The process is

(1) adiabatic

(2) isochoric

(3) isobaric

(4) isothermal

Ans. (1)

174. The energy equivalent of 0.5g of a substance is

(1) 4.5×10^{13} J

(2) 1.5×10^{13} J

(3) 0.5×10^{13} J

(4) 4.5×10^{16} J

Ans. (1)

175. A capillary tube of radius r is immersed in water and water rises in it to a height h. The mass of the water in the capillary is 5g. Another capillary tube of radius 2r is immersed in water. The mass of water that will rise in this tube is:

(1) 5.0 g

(2) 10.0 g

(3) 20.0 g

(4) 2.5 g

Ans. (2)

176. A screw gauge has least count of 0.01 mm and there are 50 divisions in its circular scale:

The pitch of the screw gauge is:

(1) 0.25 mm

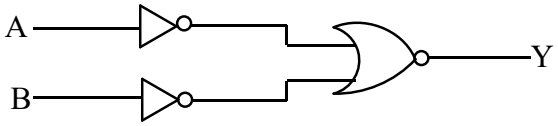
(2) 0.5 mm

(3) 1.0 mm

(4) 0.01 mm

Ans. (2)

177. For the logic circuit shown, the truth table is :



(1)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(2)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

(3)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

(4)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Ans. (4)

178. The average thermal energy for a mono-atomic gas is:

(k_B is Boltzman constant and T, absoulte temperature)

(1) $\frac{3}{2}k_BT$

(2) $\frac{5}{2}k_BT$

(3) $\frac{7}{2}k_BT$

(4) $\frac{1}{2}k_BT$

Ans. (1)

179. The solids which have the negative temperature coefficient of resistance are:

(1) insulators only

(2) semiconductors only

(3) insulators and semiconductors

(4) metals

Ans. (3)

180. Two particles of mass 5kg and 10kg respectively are attached to the two ends of a rigid rod of length 1m with negligible mass.

The centre of mass of the system from the 5kg particle is nearly at a distance of

(1) 50 cm

(2) 67 cm

(3) 80 cm

(4) 33 cm

Ans. (2)

❑ ❑ ❑

TRIPATHI STUDY ZONE
JEE MAIN | NEET | AIIMS | MHCET | BITSAT
Physics Expert
Run By. Er.JITENDRA MANI TRIPATHI
(B.Tech from Kolkata, WBJEE AIR 676)
Mob:- 9112409964